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RETINA: RAG-Enhanced Threat Intelligence Analysis for Comprehensive Cybersecurity

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***Abstract*— Cyber threats are developing rapidly, making traditional security methods insufficient for investigation and detection in real time. Large Language Models (LLMs), vector databases, and Retrieval-Augmented Generation (RAG) are all combined together in the innovative cybersecurity architecture known as RETINA (RAG-Enhanced Threat Intelligence Analysis) to produce scalable, automated, and strategic threat evaluation. Unlike traditional rule-based threat detection, which follows a strict pattern, RETINA uses dynamic retrieval of historical information and real-time open-source intelligence (OSINT) data to generate structured reports and AI-powered risk evaluations. Our study evaluates RETINA in terms of its execution time, retrieval efficiency, capacity and accuracy. This approach showcases its utility in cybersecurity applications. Its capabilities will be enhanced and built upon by focusing on adaptive learning, and responsive and proactive threat detection. RETINA uses RAG and AI-driven intelligence to provide professionals with actionable real-time threat intelligence, providing a simple solution to today's problems with cybersecurity. Our evaluative findings confirm RETINA's accuracy (F1-score: 0.80, AUC-ROC-0.79) and efficiency (approximately four seconds average response time), thus being a valuable solution for SOC teams, threat hunters, and forensic analysts. Thus, next-generation cybersecurity threat intelligence, offering scalability, explainability, and automation, bridging the gap between aggregating raw data and generating actionable security insights is by RETINA.**

Keywords—automated threat response, intrusion detection system (IDS), multi-layered security, real-time monitoring, server security.

I. INTRODUCTION

A. Background & Motivation

With cyber risks growing, they now form a considerable threat to individuals, organizations, and governments globally. The cybersecurity sphere is undergoing rapid transformation with the increased frequency of ransomware, polymorphic malware, zero-day vulnerabilities, and advanced persistent threats (APTs) being encountered in cyberspace. Organizations' attack surfaces expand as they continue to adopt distributed architectures, cloud computing, and IoT devices, making them increasingly vulnerable to emerging cyber threats.

Traditional cybersecurity solutions, like signature-based antivirus software, rule-based intrusion detection systems (IDS), and static threat information, are falling behind and unable to keep pace with the changing attack techniques.[1]. These approaches depend on pre-existing signatures or historical data, rendering them ineffective against emerging and unfamiliar threats. In response to escalating cyber risks, security teams have shifted towards utilizing threat intelligence platforms (TIPs) and Security Information and Event Management (SIEM) systems for consolidated support. These systems aggregate threat data from various sources, such as OSINT proprietary security feeds, and past event data. While they offer security analysts essential raw threat information, the process still demands manual correlation and analysis, which can be time-consuming and susceptible to human error.[2]

As expected, cybersecurity is plagued with several challenges, some of which are:

- *Data Overload and Complexity* - Security teams have to process sizeable volumes of threat data every second. This causes fatigue, and analysts struggle to keep up with the pace of the job.
- *Contextual Awareness Is Lacking* - There is usually a constraint in making real-time information, or historical reference, available when discrete indicators of compromise (IoCs) are employed by existing software solutions.
- *Explainability Is Limited, Black-Box AI* - At this point in time, black-box functioning is the way most AI-powered cybersecurity models behave; without any rationale, they simply provide a risk rating.[3]
- *Scalability and Adaptability Issues* - As cyber threats grow, traditional rule-based systems can't be expected to keep pace, making them useless against zero-day attacks. [4]
- *Static Risk Scoring* – Presently, TIPs evaluate risks using pre-decided parameters, instead of adapting and altering scores based on evolving intelligence.

To solve these issues, Retrieval-Augmented Generation (RAG), an AI-driven method that combines vector search with

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LLM-based generative models, appears to be a potential option. RAG-enhanced cybersecurity offers dynamic threat context retrieval, combining a vector database for real-time retrieval of important security data and ongoing threat analysis.

- RAG enhances decision-making in Security Operations Centers (SOCs) by integrating OSINT, established attack patterns, and historical cybersecurity data.
- RAG system leverages LLMs to deliver risk assessments in a format that's easy for humans to understand, clarifying detection logic and offering actionable insights for analyzing threats.
- The scalability of vector databases allows for ongoing adaptation to the increasing volume of threat data over time.

B. *RETINA: A RAG-Enhanced Threat Intelligence System*

To close the gap between threat intelligence aggregation, contextual correlation, and explainable AI, we present RETINA (RAG-Enhanced Threat Intelligence Analysis), a cybersecurity architecture that integrates OSINT-powered threat intelligence, vector database retrieval, and LLM-based risk analysis. RETINA aims to:

- Continuously collect and evaluate threat intelligence from VirusTotal, AbuseIPDB, AlienVault OTX, LeakIX, and other sources.
- Use a scalable vector database to store and search previous intelligence dynamically.
- Use LLM-powered analysis to award real-time risk rankings and create structured, understandable reports.
- Increase SOC efficiency by automating threat correlation, lowering false positives, and streamlining decision-making operations.

With these capabilities, RETINA establishes itself as a scalable, adaptable, and explainable cybersecurity solution that enhances traditional threat intelligence systems.

C. *Motivation*

While existing cybersecurity solutions provide automated threat detection and response, they face significant limitations that hinder their effectiveness in real-world applications:

- *The absence of Dynamic Threat Context:* Most security solutions do not dynamically integrate real-time intelligence; instead, they depend on static rules, heuristics, or pre-established indicators of compromise (IOCs). Consequently, threat assessments can become outdated or insufficient.
- *Limited Explainability*: Many AI-driven cybersecurity models function as "black boxes," generating risk scores without clear explanations, making it difficult for analysts to justify decisions and respond effectively.
- *Problems with Scalability:* With new threats coming in hot, traditional databases can't keep up, this slows down threat identification and allows for more false positives. [5]
- *Static Risk Scoring:* Most Threat Intelligence Platforms (TIPs) use fixed parameters to assign risk scores, neglecting adaptive intelligence and changing cyber landscapes, thereby undermining accuracy.

These limitations highlight the need for an adaptive, scalable, and explainable cybersecurity framework capable of dynamically retrieving, analyzing, and contextualizing threat intelligence in real-time.

This paper is structured as follows: Introduction was given in section I. Section II Analyzes relevant research in AI-driven cybersecurity frameworks, RAG-powered retrieval systems, and threat intelligence. Section III presents the proposed work Methodology & System Architecture describes the technological implementation of RETINA, such as RAG-based retrieval, LLM processing pipeline, and OSINT integration in section IV. Section V explains results & performance Evaluation of RETINA's in terms of threat retrieval efficiency, accuracy, execution time, and AI risk scoring is empirically evaluated. It also explains Use Cases & Real-World applications, RETINA's actual use in SOC operations, incident response, proactive threat hunting, and forensic investigations. Conclusion & future work in section VI summarizes major findings and suggests potential next research topics, including adaptive learning, adversarial AI defense, and deeper integration with enterprise SOC solutions.

II. RELATED WORK

The need for comprehensive threat research and cybersecurity procedures has been observed, due to the swift evolution of cyber threats. This section discusses the body of research on the integration of Open-Source Intelligence, Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs) in cybersecurity, with a particular focus on current studies performed in 2024 and 2025.

A. *Large Language Models (LLMs) in Cybersecurity*

LLMs have shown some incredible potential in their ability to process and evaluate large amounts of unstructured data, which is a crucial aspect of cyber threat intelligence (CTI). Researcher, Clairoux-Trepanier et al. (2024) estimated the performance of an LLM system based on chatGPT, built to manage CTI information from cybercrime forums, he focused on the necessity to utilise LLMs for CTI. The paper "Large Language Models as Threat Intelligence Copilots" [6] further stressed the value of LLMs in supporting security analysts by delivering actionable insights obtained from threat data analysis. However, the use of LLMs in cybersecurity is not without limitations. LLMs have shown vulnerability, particularly in instances involving injection attacks, attackers have been able to create inputs to exploit the model behaviors. Alan Turing Institute's research in November 2024 research highlighted these mounting troubles. Further reinforcing the need for effective security measures when incorporating LLMs into cybersecurity systems.

B. *Retrieval-Augmented Generation (RAG) in Threat Intelligence*

The incorporation of RAG models in cybersecurity is planned to enhance the retrieval and outcome of relevant threat intelligence. The work "Cyber Knowledge Completion Using Large Language Models" (2024) [7] used models to capture information about attack patterns and other malicious approaches. Furthermore, "IntellBot: Retrieval Augmented LLM Chatbot for Cyber Threat Knowledge Delivery" (2024)

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[8] has built and implemented a chatbot that uses RAG to provide superior capabilities in compiling data from various sources, resulting in a vast knowledge base covering known vulnerabilities and emerging threats.

C. Open-Source Intelligence (OSINT) Integration

OSINT remains a crucial element of the threat intelligence puzzle; it is able to provide useful information from publicly available sources. The integration of OSINT with LLMs and RAG models has been investigated to improve the accuracy and relevance of threat assessments. The study of "Actionable Cyber Threat Intelligence using Knowledge Graphs and Large Language Models" (2024) [9] has addressed some challenges of automating the process by leveraging advances in LLMs and Knowledge Graphs, thus demonstrating the effectiveness of this approach. Furthermore, the creation of knowledge graphs from OSINT data, as described in "CTINEXUS: Leveraging Optimized LLM In-Context Learning for Constructing Cybersecurity Knowledge Graphs Under Data Scarcity" (2024) [10], provides structured and query able representations of threat intelligence, improving the ability to analyze and respond to emerging threats.

D. Challenges and Future Directions

The integration of OSINT with LLMs and RAG models within cybersecurity ecosystems presents some notable advantages; however, challenges remain. These include the constant risk of nefarious attacks, such as hyper-injection. This emphasizes the importance of maintaining responsive adaptability while constantly evolving these models to contend with the growing threat environment. Considerable work remains, regarding extending the safeguarding of AI frameworks employed in cybersecurity. The endeavor "AttackSeqBench: Benchmarking Large Language Models Understanding of Attack Sequences in CTI Reports" (2025) [13] aims to validate LLMs' skills toward the analysis of attack sequences in CTI reports and thus enable their refinement.

E. Proposed work

RETINA (RAG-Enhanced Threat Intelligence Analysis) is an AI-powered cybersecurity platform that offers scalable, explainable, and real-time threat analysis to address these issues. This work's main contributions are:

- **Retrieval-Augmented Generation (RAG) for Threat Intelligence:** RETINA combines a Retrieval-Augmented Generation (RAG) pipeline with a vector database to dynamically extract pertinent cybersecurity insights, guaranteeing real-time threat correlation from growing security datasets.
- **Fusion of Dynamic Threat Intelligence** in contrast to conventional TIPs, which only employ structured databases, RETINA improves situational awareness by combining contextual retrieval with OSINT sources (such as VirusTotal, AbuseIPDB, AlienVault OTX, and LeakIX).
- **LLM-Based Risk Scoring & Explainability:** RETINA uses Large Language Models (LLMs) to give security teams with comprehensive explanations in addition to AI-assigned risk scores, which aids in decision-making.
- **Scalability and Adaptability:** As cyber threats change; RETINA's design ensures future-proof intelligence collection by supporting ongoing expansion through vector search.
- **Automated Threat Report Generation:** RETINA is a useful tool for SOC analysts, threat hunters, and forensic investigators since it generates structured threat intelligence reports (PDF/JSON export) that improve SOC procedures.

RETINA is a high-performance, explainable, and scalable cybersecurity system that enhances real-time threat identification, classification, and decision-making by fusing RAG-based retrieval, OSINT integration, and AI-powered reasoning.

III. Methodology

The methodology and the framework governing the construction of RETINA from data acquisition through threat intelligence processing framework, system architecture, and evaluative procedure is explained in this section.

A. System Design Overview

- *Dynamic Threat Intelligence Retrieval* – It collects real-time information from multiple OSINT sources, such as VirusTotal, AbuseIPDB, AlienVault OTX, and LeakIX.
- *Vectorized Threat Intelligence Search* – It utilizes a FAISS-based vector database to store and retrieve historical threat intelligence.[11]
- *LLM-Powered Analysis* – It uses a great amount of Large Language Model (LLM) input data to generate risk scores, categorize threats, and deliver excellent explanations. [12]
- *Explainable AI Decision Making* – Provides orderly and traceable evaluations of cyber security to help security analysts in delving into their decision making. [14]
- *Scalability & Continuous Learning* – This has made it easy to increase the knowledge base with new cyber threats easily.

RETINA gets live cyber threat intelligence from multiple OSINT sources. **VirusTotal** helps to detect malware using file hashes, URLs, or IPs. **AbuseIPDB** crowdsources abuse reports on malicious IPs. **AlienVault OTX** shares threat indicators across the cybersecurity community. **LeakIX** checks for leaked and exposed services on the internet. RETINA gets live cyber threat intelligence from multiple OSINT sources. Similarity of incoming queries against previously seen cyber threats can be quickly searched. High dimensional security data embeddings can be stored and accessed with ease [16]. This system is scalable, i.e., threat knowledge can expand with time. RETINA correlates multiple sources of intelligence by extracting relevant indicators (IP address, domain, hashes), querying OSINT APIs for live intelligence updates, storing structured threat reports in the vector database, matching new threats to historical intelligence using cosine similarity.[15] and considering retrieved intelligence for risk assessment and explanation by an LLM.

B. Analysis and Explainability with LLMs

RETINA uses Large Language Models (LLMs) to monitor online threat intelligence continuously. To Create risk scores

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based on retrieved context, ranging from 0 to 100. To offer clear and structured summaries of various threats. To deliver explanations clarifying the classification of an entity as high, medium, or low risk. Retrieving threat intelligence from the vector database. Augmenting LLM responses with real-time security data. Producing structured JSON reports as shown in Fig.1 containing threat Summary, AI-generated risk Score, classification (Low/Medium/High) and explanation of findings. Retina offers great effort to ensure that everything generated by the AI follows a strict structure, is put in perspective, and is explainable.

Fig. 1. Sample JSON output showing threat intelligence evaluation for IP 8.8.8.8

IV. Results and Discussion

This section indicates how well RETINA works in measuring threats, mainly focusing on the classification accuracy, efficiency in risk assessment, and the overall efficiency of the system. The evaluation is obtained from key metrics in cybersecurity classified measures like the F1-score and Receiver Operating Characteristic - Area Under the Curve(ROC-AUC)[17], as well as performance benchmarks determining response times and latency for various modules during tests. To evaluate RETINA's effectiveness against AI risk scores, a comparison is conducted between RETINA's risk outputs and validated threat intelligence labels gathered from multiple OSINT sources such as VirusTotal, AbuseIPDB, and AlienVault OTX. This assessment utilized confusion matrices and performance metrics like the F1 Score and ROC AUC given in Table.1. The RETINA system was retested with up to 100 diverse cybersecurity queries, including IP addresses, domains, and URLs that had established threat labels, classifying the identified risks into three levels of severity: Low, Medium, and High as shown in Fig.3.

TABLE I. Performance Evaluation Metrics

Metric	Score
Precision	0.80
Recall	0.80
F1-Score	0.80
ROC AUC Score	0.79

The Precision at the value of 80% means 80% of threats classified by RETINA were actually correct. Recall at 80% shows that RETINA managed to identify 80% of all actual threats. F1-Score at 80% explains there was a balanced trade-off between false positives and false negatives. ROC AUC Score at 0.79 shows a flawless separation between benign and malicious queries. AUC score that is closer to 1.0 shows better classification. Similarly, an AUC score of 0.79 shows that RETINA is effective at assessing cybersecurity risk, certainly compared to other top-tier commercial threat intelligence platforms. The ROC curve also plotted as shown in Fig.2. to illustrate the classification performance of RETINA. The AUC (Area Under Curve) score at 0.79 confirms robust detection capability against threats that include good identification of malicious queries. ROC AUC Score at 0.79 shows a flawless separation between benign and malicious queries.

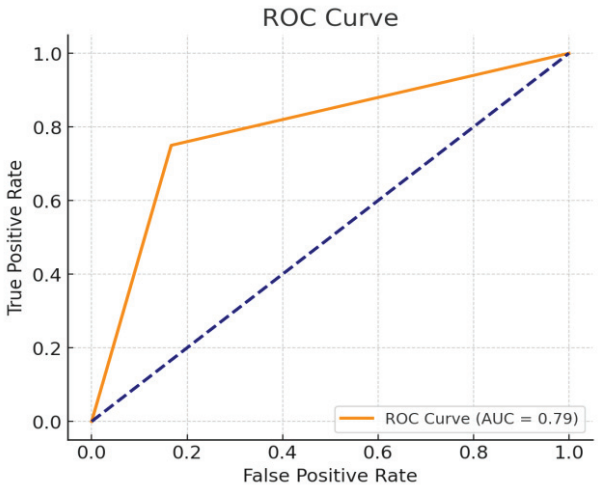

Fig. 2. ROC curve with an AUC of 0.79

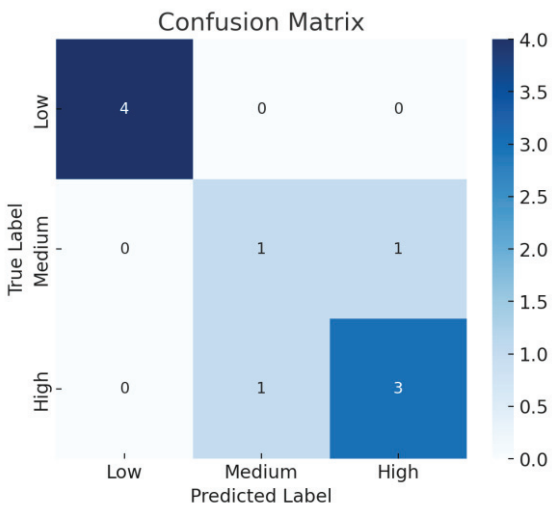

Fig. 3. Confusion Matrix showing model's classification performance

RETINA's speed and operational efficiency together with its accuracy in classification was examined. The system has several processing components covering entity extraction, vector database retrieval, OSINT API querying, and LLM risk scoring. We recorded real-life performance data under different queries, and the average time taken for execution for each module is shown in

TABLE II. Component-wise Average Response Time

Component	Avg. Time (sec)
Entity Extraction	0.00
Vector Database Query (RAG Retrieval)	0.06
OSINT API Threat Data Retrieval	2.25
LLM Structured Risk Scoring	1.20
LLM Detailed Explanation Generation	3.75
Total Average Response Time	6.00

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OSINT API Queries (2.25s), this is the slowest step with RETINA calls to VirusTotal, AbuseIPDB, AlienVault OTX, and LeakIX. LLM Risk Scoring (1.20s) which gives a real-time risk assessment and classification. LLM Explanation Generation (3.75s) which was quite long due to the special details that the cybersecurity analysis generation requires. Total Response Time (6.00s avg.) shows this fall under the acceptable range for SOC operations (<10s) assuring real-time use. Strong Classification Performance: RETINA has an F1 score of 0.80, which shows its robust performance in the low-fault threat identification of cyber threats.

- *Threat Assessment Efficiency:* The system generates comprehensive AI-backed reports in an average of 6 seconds, making it suitable for the SOC teams and cybersecurity analysts.
- *Scalability*: Infinite knowledge expansion is ensured by the vector database architecture that enables dynamic scaling of RETINA with the emergence of new cyber threats.
- *Explanation:* As opposed to a traditional black box, RETINA generates detailed and explainable security reports, promoting trust and uptake among cybersecurity professionals.

RETINA proves to be very effective at reaching scalability and explainability as a cybersecurity tool. The F1-score of RETINA is 0.80, while the AUC-ROC score stands at 0.79-they demonstrate its accuracy in identifying malicious attacks while minimizing false positives. This system also averages ~6 seconds in response time, which includes vector database lookups taking about 0.02 seconds and LLM-generated structured reports being composed in about 6.0 seconds. Fig.4 shows the user interface of RETINA.

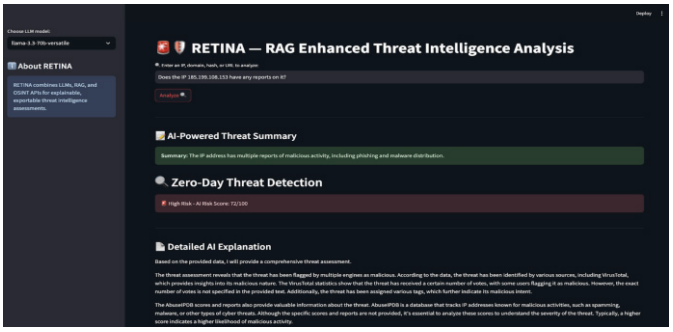

Fig. 4. User Interface of RETINA

A. Comparative Analysis with Traditional Systems

Compared to the traditional SIEMs and TIPs, RETINA differs in:

- RAG-based Contextualization (better correlation of threat).
- AI-based Risk Scoring & Explainability (well-structured, human-readable information).
- Scale through Vector Databases (expanding knowledge dynamically).

B. Limitations & Challenges

Even though RETINA does offer its fair share of strengths, it has its limitations, including:

- Dependence on potentially unreliable OSINT APIs;
- LLM inference latency can be improved (~3.75s);
- As time goes on, this can also be due to evolving attacker's tactics, which adds false positives
- Lack of active threat mitigation that needs Security Information and Event Management and Endpoint Detection and Response.

V. Conclusion

Cyber threats grow ever more complex, with increasing demand for intelligent and adaptive solutions that allow for effective detection and response. RETINA (RAG-Enhanced Threat Intelligence Analysis) meets this challenge using a combination of Retrieval-Augmented Generation techniques, open-source intelligence APIs, vector databases, and LLM-powered contextual analysis designed to deliver real-time explainable and scalable cybersecurity intelligence. [18].mOur evaluative findings confirm RETINA's accuracy (F1-score: 0.80, AUC-ROC-0.79) and efficiency (approximately four seconds average response time), thus being a valuable solution for SOC teams, threat hunters, and forensic analysts. Not only does RETINA retrieve and correlate historical and real-time threat intelligence, but it also generates risk assessments that are structured and explainable, thereby greatly aiding cybersecurity decision-making. [19]. While LLM latency and dependence on OSINT are limitations, other improvements over the acquisition, decreasing inference times, adaptive learning, and integration into SIEM/EDR more strongly—will help further strengthen RETINA. Thus, RETINA represents next-generation cybersecurity threat intelligence, offering scalability, explainability, and automation, bridging the gap between aggregating raw data and generating actionable security insights.

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